

WAUKEGAN HARBOR, IL, USA

WMO: 997735

Lat: 42.361N

Lon: 87.813W

Elev: 188

StdP: 99.09

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.7	(c) -16.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.1	(k) -0.4	(l) 12.8	(m) -1.5	(n) 6.4	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.5	(c) 30.4	(d) N/A	(e) 28.7	(f) N/A	(g) 26.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.6	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.8	(b) 12.0	(c) 10.9	(d) DB	(e) -21.5	(f) 33.1	(g) 3.5	(h) 2.4	(i) -24.1	(j) 34.8	(k) -26.2	(l) 36.2	(m) -28.2	(n) 37.5	(o) -30.7	(p) 39.3	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-4.9	-4.2	1.9	7.4	12.9	18.6	22.0	21.6	18.0	11.8	5.3	-1.0
	DBStd	10.46	6.47	5.75	4.97	4.02	4.30	3.96	3.42	2.82	3.80	4.11	5.18	5.51
	HDD10.0	1762	462	398	258	101	18	0	0	0	1	29	153	342
	HDD18.3	3673	721	632	511	329	177	44	8	7	51	203	390	599
	CDD10.0	1465	0	0	6	22	108	259	373	359	239	86	13	1
	CDD18.3	334	0	0	0	1	8	54	122	107	40	2	0	0
	CDH23.3	1723	0	0	1	6	61	311	744	437	157	7	0	0
	CDH26.7	421	0	0	0	0	11	80	217	82	32	1	0	0
(14) Wind	WSAvg	5.2	6.0	5.8	5.7	6.0	5.2	4.4	4.1	4.1	4.7	5.3	5.9	5.7
(15) Precipitation	PrecAvg	913	48	41	57	96	101	105	90	99	81	75	69	51
	PrecMax	1130	94	118	112	193	213	247	184	288	242	175	149	114
	PrecMin	599	12	5	17	43	15	25	34	49	17	19	10	20
	PrecStd	123	22	25	29	41	46	55	40	48	44	45	37	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.0	19.6	24.0	28.2	31.6	34.4	31.2	30.2	24.1	18.5	13.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	8.8	8.1	15.2	20.4	25.2	29.1	30.8	29.2	26.9	21.4	16.1	9.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	5.8	4.9	11.5	16.7	22.7	26.9	29.0	27.0	25.0	19.3	13.9	7.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	3.5	2.5	8.3	13.2	20.2	25.0	27.2	25.7	23.4	17.8	12.0	5.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	MDBR		7.1	6.6	6.0	6.5	7.4	7.0	6.5	6.0	6.4	6.6	6.4	6.2
	5% DB	MCDBR	9.1	8.5	11.4	12.7	12.3	10.7	9.3	8.0	8.7	8.8	8.6	7.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.294	0.309	0.343	0.397	0.423	0.439	0.439	0.433	0.404	0.353	0.313	0.294
	taud		2.453	2.406	2.399	2.276	2.257	2.262	2.260	2.277	2.329	2.471	2.523	2.490
	Ebn at Noon		854	895	902	871	854	839	833	826	825	835	827	823
	Edh at Noon		75	92	105	128	134	134	133	127	112	86	70	67
(44) All-Sky Solar Radiation	RadAvg		1.49	2.36	3.54	4.60	5.27	6.05	6.13	5.31	4.33	2.81	1.75	1.23
	RadStd		0.19	0.29	0.29	0.38	0.41	0.40	0.28	0.32	0.37	0.32	0.18	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page